APA Reference
Mbise, K. (2021). The Role of IT Professional Certifications in IT Instructors’ Teaching Quality. *International Journal of Education and Development Using Information and Communication Technology (IJEDICT)*, *17*, 176–187.

Summary
“This study investigates the role of information technology (IT) professional certifications in IT instructors' teaching quality in higher education institutions (HEIs) in Tanzania. The study was conducted at the Institute of Accountancy Arusha (IAA) involving IT academic staff with at least one IT professional certification. The study used a case study research design whereby mixed methods for data collection and analysis were employed. A questionnaire was used as the main data collection method supplemented by an interview. Quantitative data were analyzed using descriptive statistics. Qualitative data were analyzed using qualitative content analysis. The results of the study suggest that IT professional certifications play an important role in IT instructors' quality of teaching and the certifications improved teaching quality, raised self-confidence, kept professional skills updated, improved preparation of lesson plans and instructional materials, enhanced delivery techniques, and improved setting of competency-based learning activities. The results also suggest that the availability of certified IT instructors could have an impact on improved student performance in courses” (p. 176). “The results suggest that professional certification in the area have a positive impact on IT instructors' teaching quality. Further, IT professional certifications have the potential to improve teaching quality, raise self-confidence, keep professional skills updated, and improve the preparation of lesson plans and instructional materials. The study also demonstrates that the certifications can enhance delivery techniques and improve the setting of competency-based learning activities” (p. 184).

Method and limitations
“The targeted population was IT instructors with at least one IT professional certification teaching in computer science and IT programmes in the IAA's Department of Informatics. The study used a target population as the sample because the number of IT instructors in the department was very small, in keeping with earlier work (Kothari, 2004; Martínez-Mesa, González-Chica, Duquia, Bonamigo, & Bastos, 2016). In addition, given the importance of this study and the easy accessibility of respondents, the researcher set the margin of error (e) as zero (0)” (p. 179). “While the findings of this study provide appropriate details of the role IT professional certifications in IT instructors' teaching quality in HEIs in Tanzania and in other areas with a similar context, the findings of this study are based on one institution using a very small sample and the results may not be generalizable to a wider population” (p. 184).

How this relates to my topic and other things I’ve read? or What does this mean for my research?
Student satisfaction with the quality of a learning unit is directly related with the instructor/lecturer experience, especially with technology. The study focused on measuring the correlation of IT certifications with teaching quality. Since technology and it’s practice is advancing and spreading at a high rate, newer students are seeking more practical and valuable technology skills, especially since they are able to obtain practical experience, employment, and incomine directly from technology vendors.

Other articles I need to locate and review:
Andersson, C., Johansson, P., & Waldenström, N. 2011.
Kusumawardhani, P. N. 2017.

Topic(s):
- Information technology
- Technology certifications
- Teaching skills

Keyword(s):
- Teacher effectiveness

Database/ index/ source(s):
- Education Resources Information Center (ERIC)

APA Reference

Milovich, M., Jr, Nicholson, J., & Nicholson, D. (2020). Teaching Tip: Applied Learning of Emerging Technology: Using Business-Relevant Examples of Blockchain. *Journal of Information Systems Education*, *31*(3), 187–195.

Summary

"Within the ever-changing technology and business landscape, it is imperative that students develop skills in identifying and leveraging emerging technologies to create business value in innovative and novel ways. Drawing on the Net-enabled Business Innovation Cycle framework, applied learning techniques, and current events, we developed an assignment to explore one such emerging technology – blockchain – to enhance students' ability to apply what they have learned to solve business problems. Our findings showed that students overwhelmingly found the activity and experience beneficial in three important ways: (1) understanding an emerging technology, (2) applying the technology to contemporary business issues, and (3) leveraging what they learned to create plausible solutions to business challenges and opportunities" (p. 187). "The students demonstrated the application of their newfound knowledge by introducing innovative ways that blockchain technology might be used in cross-industry applications. Further, the students' evaluations indicated that the assignment increased their knowledge of the technology and their belief that applying the technology to a current event helped them to extend the use of the technology to other industries and/or organizations" (p. 192).

Method and limitations

"To evaluate the design of our assignment, we collected data through three methods: (1) responses to a survey, (2) unsolicited feedback about the assignment, and (3) the innovative ways in which students applied their new learning about blockchain to solve a business challenge. We collected data from 80 undergraduate students, including 1 freshman, 34 sophomores, 37 juniors, and 8 seniors who were enrolled across multiple sections of an introductory MIS course" (p. 190). "With the number of technology-related current events, this assignment could be adapted for future assignments that contribute to information systems education by introducing questions about topics that include cybersecurity, artificial intelligence, machine learning, and 5G networks" (p. 192).

Topic(s):

- Applied Learning

Keyword(s):

- Emerging technology

Database/ index/ source(s):

- Education Resources Information Center (ERIC)

How this relates to my topic and other things I've read? or What does this mean for my research?

The referenced study identifies the importance of allowing newer technology to create business value, and increase the value of the unit to the learner. With the use case limited to blockchain technology, the study does not address many technical fields, but having practical examples to provide the evidence of value increased the measure of applied learning

Other articles I need to locate and review:

LuxTag. (2019).

APA Reference
Woods, D. (2018). Introducing the Cloud in an Introductory IT Course. *Information Systems Education Journal (ISEDJ)*, *1*, 16.

Summary
"This paper discusses an effort to update an introductory IT course to add content on cloud computing. Specifically the added content was intended to provide students with experience working with cloud based infrastructure and also show how cloud offerings affect strategic aspects of IT decision making. The AWS Educate program was used to in course assignments and is also evaluated" (p. 13).
"The main conclusions that can be drawn from the assessment of the cloud content and assignments piloted in this course are that students saw the value in learning about cloud computing and found the specific discussions and assignments helpful in learning about cloud computing. The content clearly helped students understand more of the "magic" behind software and services they use on a daily basis" (p. 19).

Method and limitations
"The course activities that introduced students to cloud concepts occurred towards the end of the course. This allowed the cloud computing material to build on previous material including an introduction to problem solving using the Python programming language and an introduction to the Linux operating system" (p. 16).
"At the end of the term, a reflective assignment was used to get student feedback on the cloud content and assignments used in the course. The goal of the reflective assignment was to get student input on whether they found the content useful, whether the content should be included in future sections of the course, and suggestions for improving or expanding the content. The first part of the reflective assignment had four survey questions and the second part of the assignment asked student to provide a written reflection on their experience working with AWS. The survey portion of the assignment asked four questions. The first two used a 5 point Likert scale asking student to agree or disagree with the statements: • I found the discussion and activities related to Amazon Web Services (AWS) helpful in learning about the cloud computing technology. • I saw the value of discussing and working with AWS" (p. 18).

Topic(s):
- IT education

Keyword(s):
- Cloud computing
- IT Course

Database/ index/ source(s):
- Education Resources Information Center (ERIC)

How this relates to my topic and other things I've read? or What does this mean for my research?
This research describes the introduction of industry relevant skills to an IT course to update the content and in doing so, provide more technical awareness to students. Using Amazon Web Services to augment the experience had a positive effect on learner satisfaction.

Other articles I need to locate and review:
Mills, T. (2011).
Kelleher, K. (2015).

APA Reference
Salah, K., Hammoud, M., & Zeadally, S. (2015). Teaching Cybersecurity Using the Cloud. *IEEE Transactions on Learning Technologies*, *8*(4), 383–392.

Summary
“Effective pedagogy and delivery of cybersecurity material require not only theoretical learning outcomes, but more importantly practical and useful hands-on experience. For instance, in industry, students are not only expected to conceive the theories behind any emerging cybersecurity problem but further to demonstrate the ability of applying such theories and, consequently, design, develop, and implement innovative pertaining solutions. To provide students with such a capability, we promote using hands-on, cloud-based cybersecurity laboratories, which involve common security tools, packages, and software that are becoming applicable in cloud computing environments” (p. 383). “To summarize, using the cloud for conducting cybersecurity assignments: (1) allows instructors to easily create only once and re-use over many times preconfigured AMIs with necessary security tools and software packages, (2) enables students to rapidly and elastically allocate resources and provision EC2 instances, (3) provides a contained and safe environment, whereby instances cannot affect production networks whatsoever, (4) makes websites (which are typically blocked by governments and universities, yet are essential for teaching cybersecurity) available, (5) grants instructors the ability to monitor, troubleshoot and grade students’ work remotely and directly within their instances, and (6) allows precluding costly operations, maintenance and scheduling of physical laboratories, to mention a few” (p. 391).

Method and limitations
“In this approach, one account (i.e., the master account) is set up by an instructor using her/his own credit card. The instructor can add any credit dollar amount she/he has to this account, including grants which can be obtained from the Amazon AWS Education Team” (p. 386). “As described in the previous subsection, one main disadvantage of the centralized management approach is that students can see other students’ activities, with control privileges to stop and run their instances. To overcome this limitation, students can be asked to create their own separate accounts using their credit cards.1 This way, activities of students are not visible to each other; yet every student will pay using her/his own credit card” (p. 387).

How this relates to my topic and other things I’ve read? or What does this mean for my research?
The cybersecurity course content and the delivery using cloud resources, specifically AWS, standardize the learning experience for students as well as provide practical exposure to relevant cloud computing skills.

Other articles I need to locate and review:
F. A. Alshuwaier, A. A. Alshwaier, and A. M. Areshey. (2012)
K. Nance, B. Hay, R. Dodge, A. Seazzu, and S. Burd. (2009).

Topic(s):
- IT education

Keyword(s):
- Cloud computing

Database/ index/ source(s):
- Education Resources Information Center (ERIC)

APA Reference

CIS AND INFORMATION TECHNOLOGY CERTIFICATIONS: EDUCATION PROGRAM TRENDS AND IMPLICATIONS By. (n.d.). (2009

Summary

“Since the 1990s, Information Technology (IT) certifications have increased in number and influence throughout the information technology career field and subfields. In order to establish the validity of technical certifications, McKillip (2001) reaffirmed the findings from earlier studies and raised the question: Are relevant Information Technology industry certifications an asset to the teaching profession as they appear to be in the business world (Jenkins, 2005; Potenza, 2005; Vakhitova, 2006)?” (p. 34). “The research shows that student perceptions of instructors possessing IT certifications were favorable; instructors were found to have higher teaching effectiveness, better teaching methodology and class engagement. In addition, IT certifications permit instructors to document their domain expertise via relevant IT certifications. Furthermore, relevant IT certified instructors may be better able to deliver content because IT certifications are reflective of the real-world needs of industry. It appears that teachers with IT certification can better captivate and motivate their students” (p. 37).

Method and limitations

“Reimer's (2009) study examined student outcomes and instructor certification. Nearly 2500 students post-test results from the State of North Carolina Vocational Competency Achievement Tracking System (VoCATS) for technology education scores were collected from nine North Carolina countries to measure student achievement. To determine teacher qualifications (IT certifications), a web instrument was created and administered to 80 instructor participants in the selected countries. Potential respondents received an email notice and description of the study. They were given the Internet URL for responding to the questionnaire. Of the 80 teachers, 43 teachers participated in the study with nine teachers holding IT industry certifications. ” (p. 36).

“However, the researchers found that CIS/IT students were keenly aware of instructors who were certified. Reimers' study (2009) found a significant difference in the learning outcomes of secondary students between technology courses taught by certified and non-certified instructors; students whose instructors held IT industry certifications had higher levels of achievement than their non-certified peers (Figure 1). Andersson's (2009) study found that college undergraduate students showed a significantly greater perception of their instructor's effectiveness, teaching skills, technical expertise, and their own engagement in their classes with certified instructors” (p. 36-37).

How this relates to my topic and other things I’ve read? or What does this mean for my research?

The certification of instructors, namely industry recognized certifications, seems to correlate the satisfaction and quality of learning in technology-oriented courses. Engagement, effectiveness, and perception of expertise all seem to correlate to learner achievement when compared to non-certified instructor led courses.

Other articles I need to locate and review:

Reimers, K.W. (2009).

Kemp, P.R., & O'Keefe, R.D. (2003).

Topic(s):

- Technology certifications

Keyword(s):

- IT programs
- Teacher effectiveness

Database/ index/ source(s):

- Education Resources Information Center (ERIC)

Topic(s):
- Information Technology
- Technology Certifications

Keyword(s):
- Computing
- Industry Certifications

Database/ index/ source(s):
- Education Resources Information Center (ERIC)

APA Reference

Mardis, Marcia A.; Ma, Jinxuan; Jones, Faye R.; Ambavarapu, Chandrahasa R.; Kelleher, Heather M.; Spears, Laura I.; McClure, Charles R. (2018). Assessing Alignment between Information Technology Educational Opportunities, Professional Requirements, and Industry Demands. *Education and Information Technologies, v23 n4, 1547-1584*.

Summary

"The Florida Commission on Higher Education Access and Educational Attainment's's (2013) Aligning Workforce and Higher Education for Florida's Future report identified critical gaps between Bachelor's degree graduates and future workforce needs in the state of Florida. The authors of the gap analysis identified computing and information technology (IT) occupations as a critical area and specifically recommended that state universities and colleges focus preparing graduates to take on roles as computer network architects, computer systems analysts, computer programmers, applications software developers, systems software developers, and graphic designers" (p. 1548). "We conclude that state college and university curricula not only reflect national curriculum standards, but also prepare students to pass the examinations linked to several desired industry certifications. When internships postings and job postings were considered in light of curricula and national standards, it appeared very likely that many internships would function as a dynamic complement to coursework and certifications and culminate in the changing blend of technical and professional skills expressed in many job postings" (p. 1567).

Method and limitations

"Study participants include two universities located in a strong regional IT industry cluster (Florida Department of Economic Opportunity [DEO] 2016). Profiles were taken from the Carnegie Classification of Institutions of Higher Education (Indiana University Center for Postsecondary Research 2016): University A is a public doctoral granting institution in the BHigh Research Activity^ Carnegie classification with just over 10,000 students. It is a professional- dominant university with high, primarily residential undergraduate enrollment. University B is a large public doctoral granting institution in the Highest Research Activity^ Carnegie classification with over 43,000 students. It has a high graduate and undergraduate enrollment in balanced between arts & sciences and professions. The undergraduate population is primarily non-residential. Data were also provided by a large urban state college with just over 25,000 students" (p. 1553).
"A program's curriculum cannot be understood based only on a syllabus analysis. Additionally, while curriculum guidelines such as the ones developed by the ACM/IEEE and ACM/AIS are helpful in designing a comprehensive curriculum, programs in higher education must also respond to other factors such as faculty expertise, high school and two-year program offerings, and the needs of the local industry. These factors should be taken into account when examining a curriculum" (p. 1568).

How this relates to my topic and other things I've read? or What does this mean for my research?

The research completes study on higher education students and verifies the workforce value of students gaining practical skills in software, programming, and other technology roles. Emphasis on faculty expertise and industry environmental factors also affect the value of learners objectives upon completion.

Other articles I need to locate and review:

Al Rawi, A., Lansari, A., & Bouslama, F. (2005).
Rob, M. A., & Roy, A. (2013).

APA Reference

Marquardson, J., & Elnoshokaty, A. (2020). Skills, Certifications, or Degrees: What Companies Demand for Entry-level Cybersecurity Jobs. *Information Systems Education Journal (ISEDJ)*, *18*(1).

Summary

"Though people can pursue skills, certificates, and degrees together, financial and time constraints often make people focus their efforts to what they feel would help them most in their careers. Those seeking careers in information systems are reasonably asking themselves if the cost of a college degree is a worthwhile investment. In this paper, we analyze 11,938 entry-level job postings for cybersecurity jobs on Dice.com to determine required and desired qualifications. The results show that 7,177 (60%) of entry-level cybersecurity jobs require a college degree in a related field. Of those, 2,851 (24% of jobs) prefer a graduate degree. 3,406 (29% of jobs) require a certification. Structured query language (SQL), testing, Java, Excel, Oracle, consulting, and database skills are listed in 16% of jobs. The most popular certifications are "Certified Information Systems Security Professional CISSP" (listed in 4.8% of jobs), "Information Technology Infrastructure Library ITIL" (listed in 3.9% of jobs), "Security+" (listed in 2.9% of jobs), "Project Management Professional PMP" (listed in 2.8% of jobs) and "Information Assurance Technical IAT" (listed in 2.4% of jobs). The selected snapshot data show that college degrees are required for 60% of jobs--evidence that college degrees are still in high demand for the field of cybersecurity. However, employers are also looking for certifications and skills" (p. 22).

Method and limitations

"We created a crawler to collect data of entry level cybersecurity professional jobs from Dice.com which is a popular website for IT jobs. The web crawler searched cybersecurity jobs at Dice.com based on commonly used words for entry-level cybersecurity jobs like "Cybersecurity Analyst" or "Cybersecurity Architect." The crawler collected the 11,938 jobs that were available on Dice.com on June 10, 2019. We collected labeled information for each job post like title, keywords (tags such as analysis, firewall, python, security, and TCP/IP), job location, and the job description" (p. 25). "There are some limitations on the current analysis that should be noted. First, the search keywords were limited, so it is possible that some job postings or certifications were excluded from the analysis. Second, the current study does not predict the likelihood of a candidate being hired given a set of credentials. We can only assume that if two candidates meet the requirements, the candidates who meet more optional requirements (through either additional skills, certifications, or degrees) will be given preference. The current analysis cannot determine the extent to which an optional college degree will help a candidate gain employment when a degree is not required in the job posting. The extent to which optional qualifications aid in securing a job should be addressed in future research" (p. 27).

How this relates to my topic and other things I've read? or What does this mean for my research?

This research describes the introduction of industry relevant skills with the context of being an attractive employment candidate. The job listings compared to the curricula offered through college degrees illustrated the inherent value of a college degree and industry specific certifications together.

Other articles I need to locate and review:

Abel, J. R., Deitz, R., & Su, Y. (2014).
Burns, T. J., Gao, Y., Sherman, C., & Klein, S. (2018).
Jovanovic, R., Bentley, J., Stein, A., & Nikakis, C. (2006).

Topic(s):

- IT
- Technology Certifications

Keyword(s):

- Employment
- Industry Certifications

Database/ index/ source(s):

- Education Resources Information Center (ERIC)

APA Reference
Wierschem, D. & Mediavilla, F. A. M. (2018). Entry Level Technology Positions: No Degree Required. *Journal of Information Systems Education,* 29(4), 253-268.

Summary
"This research addresses concerns by exploring how employers value different forms of skill acquisition within the information technology environment defined as: academic degrees, certifications, and work experience. IT executives and HR managers surveyed give insight into how they relatively value the various sources for their new and experienced employees. " (p. 253). "IT employers value work experience significantly higher than either academic degrees or certifications. IT work experience had an average weight of 50% followed by academic degrees with 30% and certifications with 20%. This is noteworthy in light of the amount of literature focused on academic degrees and certifications" (p. 265).

Method and limitations
"In 2012, an anonymous survey was electronically distributed to a sample of 33,863 businesses. The survey was targeted to the head of the company's IT department and/or the HR recruiting manager in charge of hiring IT personnel" (p. 254). "The authors believe this research calls into question many of the assumptions made by today's academic institutions offering technology degrees and certifications. A significant amount of research has been done to identify the specific curriculum content that should be offered based on perceived desirability of employers. However, this research has failed to acknowledge the curriculum that employers are most (50%) interested in: experience" (p. 265).

How this relates to my topic and other things I've read? or What does this mean for my research?
The research attempts to isolate the specific perceived industry value of certifications, degrees, and work experience which can be viewed as a job needs assessment across a broad set of technology entry roles. The information is pertinent to my research to support findings that validate industry and learner expectations upon completion of a technology centric learning unit.

Other articles I need to locate and review:
Wierschem, D., Zhang, G., & Johnston, C. R. (2010).

Topic(s):
- IT
- Technology Certifications

Keyword(s):
- Employment skills
- Industry Certifications

Database/ index/ source(s):
- Education Resources Information Center (ERIC)

APA Reference

Messing, J., & Altas, I. (2006). Using IT Industry Practices to Modernize University Education. *Industry and Higher Education*, *20*(1), 25–30.

Summary

"Incorporating IT industry certification within the curriculum of university Masters courses has solved a number of problems associated with the relevance and currency of the content. However, there are other factors that have led to the approach used at Charles Sturt University in Australia, which has made it a world leader in this area. This paper discusses the factors that have been integral to the process. Among the most significant of these are a flexible delivery model that makes use of a long tradition in distance education, a well-developed university infrastructure to produce and deliver materials on a global scale, extensive experience in the use of information and communications technologies, the use of online examinations that borrow from the model used by IT industry certification and, most significantly, the support of commercial IT training organizations" (p. 25). "…CSU took the academically bold step of implementing a course that integrated IT certification with traditional university subjects. The result has been an outstanding academic and commercial success. It has solved some of the difficulties associated with university courses being seen as either outdated or not immediately relevant to working IT professionals. The factors discussed above have each contributed to this success in ways that may not be immediately obvious to outsiders, who might view the innovation as simply a matter of integrating curricula" (p. 30).

Method and limitations

"In July 2002 CSU was approached by a former IT certification trainer, Martin Hale, to investigate the possibility of capitalizing on the university's strength in DE (distance education) by developing a course that would integrate industry certification" (p. 27). "The first course to be offered in December 2002, the Master of Networking and System Administration, was planned with an intake of 40 students as the target" (p. 27). "As already noted, the planned recruiting schedule called for about 40 students, but demand has been such that in just over two years since the first intake there are two additional courses in operation and over 500 students enrolled each semester. The original Master of Networking and System Administration has been complemented by a Master of System Design and a Master of Information Systems Security. The success of these programmes has resulted in Microsoft approaching CSU to develop a fourth Masters course, ahead of its release of the next generation of database certification" (p. 29)

How this relates to my topic and other things I've read? or What does this mean for my research?

The study identifies the benefits and perceived value of integrating a complete industry certification track and in the introduction of certified trainers and personnel to ensure the success of the program. Furthermore, the expansion of the program into other degree curricula and interest in expansion by potential industry partners is encouraging in illustrating successful strategies technology courses.

Other articles I need to locate and review:

Davidson, P. (2004).

Topic(s):

- IT
- Technology Certifications

Keyword(s):

- Industry Certifications
- Instructional Design
- Partnerships

Database/ index/ source(s):

- Education Resources Information Center (ERIC)

APA Reference

McCaffery, R. N., Backus, L., & Maxwell, N. (2020). Embedding Industry Certifications into Community College Programs. *New Directions for Community Colleges*, *2020*(189), 53–66.

Summary

"Multiple stakeholders reap the benefits from building and structuring degree programs that also provide students with industry and professional credentials (Zanville, Proter, & Ganzglass, 2017), as the information technology (IT) industry, in particular, has shown. By interweaving certifications into IT degree programs, students can obtain industry-recognized certifications to prepare for work while they gain academic credits and credentials that will allow them to continue their education. Employers gain a pipeline of workers with needed knowledge and skills. Furthermore, by working with employers to ensure students gain the knowledge and skills needed to meet certification requirements, colleges ensure that their curriculum is up to date with industry standards. " (p. 53). "…development of the SCProTM Fundamentals credentials allowed students to "see" a framework for the industry and help them to better understand their options. The design of the program provided another benefit to students because they could participate in the program and earn the certifications even if they did not possess a college degree or had not previously attended college. Access to the content was possible in a nonacademic setting, and students could earn one or more credentials" (p. 63).

Method and limitations

"Using the eight-sector organizational model for SCM already used by CSCMP, the consortium set out to develop and align eight corresponding certification tracks. Ultimately, eight certifications were developed in the areas of SCM principles, customer service operations, transportation operations, warehousing operations, demand planning, inventory management, manufacturing and service operations, and supply management and procurement" (p. 59). "For those courses where the content very closely matched the design of the existing course, the embedding generally took the form of replacing the existing textbook with SCProTM Fundamentals content. In most cases, supplemental materials from Open Educational Resources (OER) were identified either to reinforce concepts from SCProTM Fundamentals content, to explain local applications of SCProTM Fundamentals content, or to add sufficient content to meet the course hours-of-instruction requirements" (p. 60).

How this relates to my topic and other things I've read? or What does this mean for my research?

The study showed the methodologies, challenges, and effectiveness of augmenting and in some cases, replacing course content with certification content completely. Student benefits include more attractive resumes and supplement their associate degree paths.

Other articles I need to locate and review:

Carnevale, A. P., Rose, S. J., & Hanson, A. R. (2012).

Topic(s):

- IT
- Technology Certifications

Keyword(s):

- Industry Certifications
- Partnerships

Database/ index/ source(s):

- Education Resources Information Center (ERIC)

APA Reference
Laverie, D., Humphrey, W., Manis, K. T., & Freberg, K. (2020). THE DIGITAL ERA HAS CHANGED MARKETING: A GUIDE TO USING INDUSTRY CERTIFICATIONS AND EXPLORATION OF STUDENT PERCEPTIONS OF EFFECTIVENESS. *Marketing Education Review*, *30*(1), 57–80.

Summary
"Based on our experience implementing digital credentials into courses, along with helping colleagues integrate other certifications into their courses, we believe this new pedagogical approach has many merits. First, it allows faculty teaching a new digital subject to learn the material and earn the credentials. Second, it allows students to claim a level of expertise, have it verified by a third-party industry leader, and then demonstrate the expertise through an applied service-learning project or other coursework. Third, certification in subject areas by multiple industry-leading companies is much more powerful than a grade on a transcript itself, no matter how difficult the assessment may have been. These certifications provide signals to hiring managers will recognize immediately regarding the skills learned and credentialing organizations. Fourth, this approach is appreciated by students and helps them to become lifelong learners. Fifth, the materials are updated much more quickly than a textbook. Finally, these certifications work beautifully in concert with other effective teaching strategies, such as active learning, service-learning projects, and flipped classrooms. Implementation is simple and cost-effective (most are free)" (p. 74).

Method and limitations
"The survey was administered via Qualtrics and communicated to students on the LMS and in class. One week at the end of the semester was used for data collection. The survey included questions to assess students' perceptions of the certifications, including scaled and open-ended questions. The measures were on a five-point Likerttype rating scales anchored by 5 = strongly agree and 1 = strongly disagree. Participation in the survey was optional; of the 70 students in two sections of the Digital Marketing course, 45 completed the survey results in a response rate of 64%. Respondents were 52.5% male and 47.50% female" (p. 70). "The data are from two sections of a Digital Marketing overview course; it would be beneficial in the future to conduct studies of certification use and student perceptions in individual courses that cover specific digital topics such as social media and analytics. Furthermore, the study was with undergraduates; future research should explore graduate courses as well. The sample size for the quantitative study was small. Finally, certifications will continue to emerge and evolve, so future research should continue to study the effectiveness of certifications" (p. 73-74).

How this relates to my topic and other things I've read? or What does this mean for my research?
The study concluded that industry certification integration can have a positive effect and benefit for learners. Because of the digital nature of marketing, its beneficial to see additional contexts on the challenges and benefits of technology driven certifications for learners.

Other articles I need to locate and review:
Caruana, A., La Rocca, A., & Snehota, I. (2016).
Eastman, J. K., Iyer, R., & Eastman, K. (2011).
Liaw, S. (2008).

Topic(s):
- Industry Certifications

Keyword(s):
- Digital content
- Teaching effectiveness

Database/ index/ source(s):
- Education Resources Information Center (ERIC)

APA Reference

Mohamed Hashim, M. A., Tlemsani, I., & Matthews, R. (2021). Higher education strategy in digital transformation. *Education and Information Technologies*, 27, 3171-3195.

Summary

"Digital transformation in the global higher education industry determines the future roadmap to a sustainable education management strategy. This research paper aims to develop a qualitative model that advocates how digital transformation as a propelling force could be used to build competitive advantages for universities. Building competitive advantage is a relative, evolving, and important concept in strategy formulation. In recent years, specifically in the education industry, the notion of building competitive advantage is challenged by global phenomena such as digital transformation globalization, information exchange, digitization and social media in most of the global industries" (p. 3171).

Method and limitations

"The paper synthesizes the literature on how universities can build sustainable competitive advantage by leveraging evolutionary learning, Information Technology and digital transformation capabilities. The paper also attempts to develop a fundamental and a detailed model derived from the extensive literature review. Finally, the paper contributes to developing numerous research propositions. In a holistic view, this paper shed light on the strategic importance of developing the combination of empiricist and pragmatist views in developing digital advantages but using a unique approach. Through the development and integration of numerous theoretical constructs, the paper highlights the significance of building sustainable competitive advantages for universities regulating the impactful changes within the capabilities of evolutionary learning and IT capabilities" (p. 3185-3186).

How this relates to my topic and other things I've read? or What does this mean for my research?

The study showed a very interesting strategy for universities to maintain relevance despite many alternative learning paths leaners can opt into. Almost every solution requires technology interactions through digital transformation, social media, and digitization with cloud computing, artificial intelligence, IoT, and big data.

Other articles I need to locate and review:

Koller, D. (2018)
Hancock, S. (2019).
Shaughnessy, H. (2018).

Topic(s):

- Education strategy

Keyword(s):

- Digital transformation

Database/ index/ source(s):

- Education Resources Information Center (ERIC)

APA Reference

Komljenovic, J. (2021). The rise of education rentiers: digital platforms, digital data and rents. *Learning, Media and Technology*, vol 46, 3, 320-332.

Summary

"The education sector is fast digitalising all of its operations. A large part is driven by proprietary digital products and services developed and offered by for-profit companies that form the education technology industry. This article aims to introduce a theoretical focus of rentiership and assetization into the study of the political economy of education technology. It discusses five potential transformations that the education sector is undergoing as a consequence of digital rentiership. These transformation address new rentee and potential rentier roles of education institutions, nestedness of digital platforms and their terms of use, a rise of contractual governance within the education sector, reinstitutionalising the sector, and tensions between competition and monopoly in digital education markets" (p. 320).

Method and limitations

"Digital rentiership in education is analysed through five implications. First, education institutions increasingly pay monetary rent, while their students and staff pay data rent. Digital plat- forms extend value chains in education and establish new rentier-rentee relations in the sector. Second, platforms connect to education institutions' digital infrastructure in different techno- legal ways. Three levels are identified, namely, full integration, nestedness, and external development. Each case has a different arrangement of ownership and control. However, students and staff as end-users become liable to various terms of use and data policies that include their education institution's policies as well as external platforms'. Third, these terms of use together with negotiated contracts between platform companies and education institutions represent the expansion of contractual governance in education. It remains to be studied how useful and understand- able the user consent is for end-users in practice. Fourth, unbundling the student experience, decision-making aided by algorithms and automated processes, and individual use of platforms bypassing education institutions might contribute to deinstitutionalising education. However, new rules and forms of valuation and recognition are being set up, leading to reinstitutionalising education. Finally, digital platforms' monopoly tendencies might have more considerable consequences for the sector once more platforms grow in size and power" (p. 330).

How this relates to my topic and other things I've read? or What does this mean for my research?

The article discusses several facets of the digital education industry, focusing on the economic, digital, and governance of education in order to reestablish learning norms, environments, and content.

Other articles I need to locate and review:

Castañeda, Linda, and Neil Selwyn. 2018.

Topic(s):

- Education strategy

Keyword(s):

- Digital transformation
- Rentiership

Database/ index/ source(s):

- Education Resources Information Center (ERIC)

APA Reference

Guneri Sahin, Y., & Celikkan, U. (2020). Information Technology Asymmetry and Gaps Between Higher Education Institutions and Industry. *Journal of Information Technology Education: Research*, *19*, 339–365.

Summary

"This paper investigates the gaps between industry and academia perceptions of information technology fields, such as computer science, software engineering, and computer engineering, and it identifies areas of asymmetry between curricula and industry expectations. The study mainly focuses on the skills required of IT professionals (graduated students) and on how higher education institutes equip students for industry. Higher education institutes have several IT-related departments. However, it is not clear whether these departments have sufficient content to equip students with industry-related skills. Rapid advances mean that some curriculum topics are redundant before the end of a standard two- or four-year degree programs. Balancing the technical/non-technical skills and adjusting the curricula to better prepare the students for industry is a constant demand for higher education institutions. Several studies have demonstrated that a generic curriculum is inadequate to address current IT industry needs" (p. 339).

Method and limitations

"The study involved a comprehensive survey of IT professionals and companies using a Web-based questionnaire sent directly to individual companies, academics, and employers. 64 universities and 38 companies in 24 countries were represented by the 209 participants, of whom 99 were IT professionals, 72 academics, and 38 employers" (p. 339). "The results indicate a lack of emphasis on personal and non-technical skills in undergraduate education compared to general computer science, software development, and coding courses. Employers' and software experts' responses emphasize that soft skills should not be ignored, and that, of these, analytical thinking and teamwork are the two most requested. Rather than a theoretical emphasis, courses should include hands-on projects. Rapid developments and innovations in information technologies demand that spiral and waterfall models are replaced with emerging software development models, such as Agile and Scrum development" (p. 340).

Topic(s):

- Education strategy

Keyword(s):

- Digital transformation
- Industry partnerships
- Higher education

Database/ index/ source(s):

- Education Resources Information Center (ERIC)

How this relates to my topic and other things I've read? or What does this mean for my research?

The article discusses generalizing technology-oriented learning units to make them more aligned with changing industry needs and standards. Understanding various risk mitigation strategies for the forward drift of technology to enhance the value and competitiveness of institutional tech education is essential for the research strategy for my problems of practice.

Other articles I need to locate and review:

Benamati, J. H., Ozdemir, Z. D., & Smith, H. J. (2010).
Celikkan, U., & Sahin, Y. G. (2017).

APA Reference
Spangler, D. (2019). **Micro Approach, Major Impact: With Microcredentials, Educators Can Tailor Learning to Their Specific Needs**. *The Learning Professional*, *40,* 60-64.

Topic(s):
- Education strategy

Keyword(s):
- Microlearning
- Industry training

Database/ index/ source(s):
- Education Resources Information Center (ERIC)

Summary
“Microcredentials are different from traditional professional learning approaches because they are: Competency-based: Microcredentials focus on evidence of teachers’ skills and abilities, not on the amount of seat time they’ve logged in their learning. Personalized: Teachers select microcredentials to pursue based on their own needs, their students’ challenges and strengths, school goals, district priorities, or instructional shifts. They work through specific activities that will support them in developing each competency. On demand: Microcredentials are responsive to teachers’ schedules. Educators can opt to explore new competencies or receive recognition for existing ones any time of the day, using an online system to submit evidence for evaluation. Shareable: Educators can share the learning they gained through their microcredentials with other educators within and outside their school district, thereby serving as resources and mentors for other teachers” (p. 61).

Method and limitations
“In summer 2017, I designed an inhouse microcredential pilot including the coursework, tasks, evidence requirements, and evaluation process. I spent that first summer using our school district’s learning management system to design the coursework and tasks” (p. 63). “The course, broken up into three sections, ran from September through mid-April. Each month, teachers engaged in self-paced instruction (three to five hours per month) using materials aligned to district priorities and other professional learning. Learning occurred via a mix of online and face-to-face instruction. By the end of the first section, all volunteer participants demonstrated competency through shared discussion board postings and reflection pieces submitted to administration. All participants were able to: • Explain the evaluation requirements of the course; • Identify what is and isn’t innovative in education and explain why teachers today need to innovate; • Define what a microcredential is and why it is different from traditional professional learning; • Recognize there are seven Standards for Professional Learning and identify the two standards to be developed through this course; • Explain why this course format will use Danielson’s six Framework for Teaching Clusters; and • Reflect on what personalized professional learning is and isn’t. In the second and third sections, coursework focused on macrolevel practices (knowledge and skill development) and micro-level learning (practice and classroom application) in the areas outlined in the Learning Communities and Data standards” (p. 63).

How this relates to my topic and other things I’ve read? or What does this mean for my research?
Microcredentials are more pragmatic and flexible with respect to meeting learner or industry needs. The study conducted showed noticeable effectiveness by both learners and instructors. As an educational strategy, the needs of stakeholders can be approached in an effective manner and increases he chances of a more practical learning experience.

Other articles I need to locate and review:
None

APA Reference

Hidayah, N., Ahmat, C., Arif, M., Bashir, A., Razali, A., & Kasolang, S. (2021). *Micro-Credentials in Higher Education Institutions: Challenges and Opportunities*.

Summary

"The latest revolution in online education, micro-credential, is growing interest among public and private universities worldwide, including Malaysia. However, to date, little scholarly work is found related to micro-credentials in higher education. This conceptual paper presents an overview of micro-credential and the challenges and opportunities of offering micro-credential certification in the form of digital badges to the national and global market. Recommendations are made to multiple stakeholders (e.g., higher education providers, employers) to enhance the use of certifications for graduate employability" (p. 281). "The MQA define MC as "a term that encompasses various forms of certifications, including nanodegrees, micro-masters' credentials, certificates, badges, licenses, and endorsements…As the name implies, MC focus on a much smaller volume of learning than the conventional awards, which allow learners to complete the required study over a shorter duration" (p. 282). "Based on the reviews, the MC certification programme's success depends on the various stakeholders (i.e., learners, educators, universities, MC providers, employers, regulators). Support from the Ministry of Education Malaysia and the MQA as the regulators is needed in providing guidelines and assistance to ensure the MC offered is well-accredited and employability-driven. Additionally, higher education providers and MC providers should have a clear guideline to avoid any vagueness of MC implementation processes" (p. 288).

Method and limitations

"UiTM Cawangan Pulau Pinang, one of the UiTM branch campuses, organized a seminar on 19th October 2019 to create awareness among educators and learners and spark interest among module developers. Educators who have been actively involved in developing online courses were selected as module developers and were trained to develop MC modules and learning materials. After the seminar, a series of workshops were conducted to train the module developers to unbundle the courses, design the modules, record and edit videos, and incorporate suitable assessments for each module. The Micro-Credentials@UiTMCPP project aimed to offer MC for professional and personal development of the current and potential students and working adults" (p. 63).

How this relates to my topic and other things I've read? or What does this mean for my research?

Microcredentials are more pragmatic and flexible with respect to meeting learner or industry needs. One of the strategies employed by institutions are shorter, concise learning paths which align with learner employability.

Other articles I need to locate and review:

DeMonte, J. (2017).

Topic(s):

- Education strategy

Keyword(s):

- Microlearning
- Industry training
- Digital badges

Database/ index/ source(s):

- Education Resources Information Center (ERIC)

APA Reference
Koranteng, F. N., Wiafe, I., & Kuada, E. (2018). An Empirical Study of the Relationship Between Social Networking Sites and Students' Engagement in Higher Education. *Journal of Educational Computing Research*, *57*(5), 1131–1159.

Summary
"This article investigates how students' online social networking relationships affect knowledge sharing and how the intensity of knowledge sharing enhances students' engagement. It adopts the social capital theory as the basis for investigation, and the partial least square structural equation modeling was used to examine the hypothesized model. Responses from 586 students in higher education were analyzed. The findings provided empirical evidence which contradicts the argument that students perceive social networking sites as an effective tool for learning. Also, contrary to previous studies which posit that knowledge sharing impacts engagement, it was observed that there is no relationship between the two" (p. 1131).

Method and limitations
"Self-reporting approach was used for data collection, and participation was purely voluntary. Respondents were asked to indicate their positions on a 5-point Likert-type scale on their agreement or disagreement on the various relevant issues regarding social networks use and its potential to keep them engaged in learning activities " (p. 1140). "Considering all this, the use of SNSs for academic purposes is still limited and therefore calls for investigations in different educational institutions, geographical, and sociocultural settings (Rodrı´guez-Hoyos, Salmo´n, & Ferna´ndez-Dı´az, 2015)" (p. 1134).

How this relates to my topic and other things I've read? or What does this mean for my research?
Knowledge sharing and social networking technologies and mechanisms are used by learners to share their knowledge with each other. This study attempts, unsuccessfully to correlate the effects of knowledge sharing on engagement and effectiveness.

Other articles I need to locate and review:
Waycott, J., Thompson, C., Sheard, J., & Clerehan, R. (2017).
Scott, K. S., Sorokti, K. H., & Merrell, J. D. (2016).

Topic(s):
- Learning effectiveness

Keyword(s):
- Higher education
- Knowledge sharing
- Learner engagement

Database/ index/ source(s):
- Education Resources Information Center (ERIC)

APA Reference

Arbeit, C., Bentz, A., Cataldi, E. F., & Sanders, H. (2019). *Alternative and Independent: The Universe of Technology Related "Bootcamps."*

Summary

"In recent years, nontraditional workforce training programs have proliferated inside and outside of traditional postsecondary institutions. A subset of these programs, bootcamps, advertise high job placement rates and have been hailed by policymakers as key to training skilled workers. However, few formal data exist on the number, types, prices, location, or other descriptive details of program offerings. We fill this void by studying the universe of bootcamp programs offered as of June 30, 2017. In this report, we discuss the attributes of the 1,010 technology related programs offered in the United States, Canada, and online" (p. ii). "bootcamps are typically depicted as short, intense, vocational coding and web development programs designed for students who are looking to move into high-skill technology careers through a single comprehensive program. In fact, the program type that most closely resembles the media's portrayal of a typical bootcamp—comprehensive career preparation programs—comprises only 50 percent of all programs offered by bootcamps in the US, Canada, or online. The other half of bootcamp program offerings, however, defy this depiction. About 38 percent of programs are courses, which are not intended for career changers and resemble single college courses in the amount of material presented. Seven percent of programs are offered by traditional accredited institutions but are ineligible for university credit; and about 4 percent of programs are fellowship programs or postsecondary education replacement programs, which both have intense admissions processes and are designed for individuals with specific skills. The subject areas these institutions teach extend beyond web development, also encompassing data science and information technology/security" (p. 20)

Method and limitations

"We examine bootcamps at both the bootcamp (provider) and program levels. Provider-level analyses allow us to look at the breadth of offerings at both the typical bootcamp and larger bootcamps. In addition, these analyses allow us to consider whether the industry is driven by a small number of large bootcamps or by many small bootcamps. Program level analyses allow us to quantify the overall number of program offerings and identify patterns for programs with certain characteristics" (p. 8) "…a limitation to our approach since we describe programs without the context of program size. Because bootcamps rarely publish how many students enroll in each program or how often programs are offered, we may be counting equally programs that are offered once per year for 10 students and programs that are offered 10 times per year for 20 students at a time" (p. 9).

How this relates to my topic and other things I've read? or What does this mean for my research?

The strategy of using bootcamps to accelerate learning provides insights into the perceived value of the technology skills by learners. In addition, understanding satisfaction and effectiveness of bootcamps that affiliated with higher education institutions provides insight into opportunities which can be replicated or expanded.

Other articles I need to locate and review:

Powers J. F. (2017).
McBride, S. (2016, December 6).
Thayer, K., & Ko, A. J. (2017).

Topic(s):

- Learning effectiveness

Keyword(s):

- microlearning
- bootcamps

Database/ index/ source(s):

- Education Resources Information Center (ERIC)

APA Reference

Yin, J., Goh, T.-T., Yang, B., & Xiaobin, Y. (2020). Conversation Technology With Micro-Learning: The Impact of Chatbot-Based Learning on Students' Learning Motivation and Performance. *Journal of Educational Computing Research*, 073563312095206.

Summary

"This study investigated the impact of a chatbot-based micro-learning system on students' learning motivation and performance. A quasi-experiment was conducted with 99 first-year students taking part in a basic computer course on number system conversion. The students were assigned to a traditional learning group or a chatbot based micro-learning group. After the experiment, both groups achieved a comparable performance, suggesting that students are sufficiently competent to learn independently in the chatbot-based learning environment without the need for continuous face-to-face delivery. Moreover, students in the chatbot learning group attained significantly higher intrinsic motivation than the traditional learning group with perceived choice and perceived value as core predictors of intrinsic motivation" (p. 154) "This study reveals that the chatbot-based micro-learning strategies comprise a promising technology and are effective in supporting learning in basic computer knowledge among university students. The Self Determination Theory (SDT) approach in this study is an appropriate theoretical lens to measure student motivation in chatbot based learning. The outcomes of the investigation provide substantial promise for the further development of chatbot-based micro-learning systems with different degrees of interaction" (p. 171)

Method and limitations

"The participants of this study were first-year university students in the class of 2019. A quasi-experimental design was adopted in this study as the participants were not randomly assigned (Hallberg & Eno, 2015). Participants were grouped according to their scheduled classes. The traditional learning class (the control group) consists of 48 students majoring in International Economics and Trade (17%) and Business Administration (31%) whereas the chatbot-based learning class (the experimental group) consists of 51 students majoring in Education (35%) and Tourism Management (16%). Overall, the participants consist of 91 female students and 8 male students which is a typical distribution for the business and social science faculties in this university" (p. 159)

"This study administered two groups of pre-test and post-test questionnaires to examine the effectiveness of chatbot-based micro-learning compared to traditional learning" (p. 161-162).

How this relates to my topic and other things I've read? or What does this mean for my research?

Leveraging artificial intelligence, specifically chatbots, to assert whether motivation and performance are affected is an important educational strategy to consider when attempting to increase learner engagement and satisfaction

Other articles I need to locate and review:

Aluja-Banet, T., Sancho, M.-R., & Vukic, I. (2019).
Gupta, S., & Jagannath, K. (2019).
Shail, M. S. (2019).

Topic(s):

- Learning performance

Keyword(s):

- microlearning
- chatbot
- artificial intelligence

Database/ index/ source(s):

- Education Resources Information Center (ERIC)

APA Reference
Newby, T. J., & Cheng, Z. (2019). Instructional digital badges: effective learning tools. *Educational Technology Research and Development*, 68, 1053–1067

Summary
"Digital badges, widely known as alternative or micro-credentials, have gained increasing recognition in recent years as innovative pedagogical tools in higher education. Despite many anecdotal and conceptual statements of value, the effectiveness of using digital badges to improve learning performance is still largely unknown. This study addressed this gap by investigating the impact of these badges with added instructional scaffolding on pre-service teachers' perceived technology capabilities and their actual learning performance while studying within a large undergraduate technology integration course. Compared with similar participants who experienced traditional instructional projects instead of the badges, those learning with digital badges not only reported higher levels of perceived confidence in their technology integration skills but also achieved higher levels of course assignment and overall course grades. Conclusions from this study offered ways to improve learning performance with the support of digital badge technology and drew implications for future scholarship in this area" (p. 1053)

Topic(s):
- Learning performance
- Microcredentials

Keyword(s):
- Digital learning
- Digital badge

Database/ index/ source(s):
- Education Resources Information Center (ERIC)

Method and limitations
"This study was conducted in a large land-grant Midwestern university. The total sample consisted of 150 preservice teachers, in two lecture sections of a college-level educational technology course, each with 78 and 72 students respectively. One lecture section was designated as the instructional digital badge section (DB); whereas, the other lecture section became the traditional project section (TP). Each lecture section met for 1 h each week during the 16-week course. Students in each lecture were subdivided into smaller labs (20–25 students/each) that met for one 2-h lab session each week. Each lab was conducted by a graduate teaching assistant" (p. 1058) "First, although badge and grading rubrics were matched and training was provided, there was a variety of teaching assistants assigned to the grading of the assignments. Variability between the graders could have impacted the measure of performance on the learning tasks and the overall course grade across labs…. Second, instructional digital badges as instructional interventions are still fairly novel for most learners. Thus, a novelty effect may have increased motivation and invested effort. In such cases, as familiarity is gained with such an intervention, temporary increases in motivation may be shown to be limited. From another viewpoint, a novelty effect could also limit the degree of increased perceived skill capabilities and performance gains. In such cases, as the acceptance of this innovation is gained and more supporting instructional design guidance for this intervention are developed, students might show greater gains in perceptions and learning performance in similar learning contexts. Finally, self-reported Likert-type surveys were used to measure perceived technology skills, which could have a weakness in predicting accuracy as a data collection tool" (p. 1064).

How this relates to my topic and other things I've read? or What does this mean for my research?
Digital badging and microcredentials are mechanisms which may affect learner performance and effectiveness. By shortening and focusing the design of courses, learners can obtain gratification and satisfaction with the rigor and value of the learning experience in a shorter time frame than traditional learning paths.

Other articles I need to locate and review:
Wallis, P., & Martinez, M. S. (2013).

APA Reference

Kay, J., & Smith, J. (n.d.). The emerging future: Innovative models of work-integrated learning. *International Journal of Work-Integrated Learning*, Special Issue, 2019, 20(4), 401-413.

Summary

"Work-integrated learning (WIL) is a national priority and a strategic direction for Australian universities. To increase industry engagement, there is a need to identify emerging WIL models that enable flexibility while optimizing outcomes for stakeholders. This Australian Technology Network (ATN) project explored approaches that overcome constraints to engagement, particularly for small to medium enterprises (SMEs). The preliminary qualitative phase comprised a literature review, workshops and interviews. Examples of emerging WIL models, both curricula and co-curricular, were clustered into five models: micro-placements, online projects or placements, hackathons, competitions and events, incubators/ start-ups and consulting. This paper outlines these models and summarizes defining features, enablers, challenges and opportunities. Contemporary trends informing innovative WIL design and implementation emerged. Features fell into three broad areas, stakeholder engagement, design elements and co-design with partners. Enablers, challenges and opportunities in the implementation of these WIL models that respond to the changing nature of work were documented" (p. 401)

Topic(s):

- Work-integrated learning

Keyword(s):

- Codesign
- Learning quality

Database/ index/ source(s):

- Education Resources Information Center (ERIC)

Method and limitations

"Stage one involved a search of contemporary literature, spanning the period 2012 to 2017, to identify emerging WIL models, strategies and their key features. Searches included peer-reviewed academic literature, grey literature such as government and industry reports (e.g., PhillipsKPA, 2014; AiG, 2016; Edwards, Perkins, Pearce, & Hong, 2015) and conference proceedings (e.g., ACEN, WACE) that have the advantage of shorter publication times…. Stage two involved the exploration of these emerging models with university WIL practitioners through ten international, national and local workshops, webinars and WIL communities of practice meetings. These events attracted in excess of 450 participants…. Stage three involved further in-depth consideration and analysis of the emerging models through 55 semi-structured interviews with WIL practitioners, industry partners and students who were involved directly with emerging WIL strategies. Interview participants were identified during research stages one and two and through national and international WIL communities of practice" (p. 403-404)

How this relates to my topic and other things I've read? or What does this mean for my research?

Implementing curricula which allows direct access to the skills or roles in the workforce provides learners with a contextually relevant experience that leaves very little room for drift between theory and practice. In addition, learners may perceive an authentic learning experience which may affect satisfaction and performance.

Other articles I need to locate and review:

Business-Higher Education Forum [BHEF]. (2013).
Australian Collaborative Education Network [ACEN]. (2015). N
Helyer, R., & Lee, D. (2014).
Rowe, A., & Zegwaard, K. (2017).

Topic(s):

- Work-integrated learning

Keyword(s):

- Internship
- Learning quality

Database/ index/ source(s):

- Education Resources Information Center (ERIC)

APA Reference

Jaime, A., Olarte, J. J., Garcia-Izquierdo, F. J., & Dominguez, C. (2020). The Effect of Internships on Computer Science Engineering Capstone Projects. *IEEE Transactions on Education*, *63*(1), 24–31.

Summary

"Internships designed to provide training and an initial period of contact with industry, prior to a computer science engineering capstone project, have a very positive impact on both industry and academic capstone projects" (p. 24) "Internships prior to capstone projects improve student skills in autonomy, technology, methodology, and project management; increase the complexity and technological novelty of the resulting projects; and reduce advisor involvement in practical (technology, execution) and keep-the-project-alive issues, and increase advisor involvement in monitoring student work (meetings, reports, and initial arrangements). This effect was observed in both industrial and academic projects" (p. 24).

Method and limitations

"The hypothesis was that the completion of an internship will have positive effects on several aspects of the capstone projects: 1) improved student competencies; 2) improved capstone project outcomes; and 3) decreased supervision effort. Further, these positive effects were expected to be greater in industry-based projects than in academic projects. The hypothesis was tested through a quantitative study of data collected from 274 computer science engineering capstone projects. A period of time with internships was compared with another period without internships, and differentiating between academic and industry projects" (p. 24)

How this relates to my topic and other things I've read? or What does this mean for my research?

Exposure to working environments and needs within course work, specifically a capstone project, appears to correlate to students needs for both support and independence in project related learning.

Other articles I need to locate and review:

J. R. Goldberg, V. Cariapa, G. Corliss, and K. Kaiser, (2014).
J. F. Binder, T. Baguley, C. Crook, and F. Miller, (2015).

APA Reference
Leaser, D., Jona, K., & Gallagher, S. (2020). Connecting Workplace Learning and Academic Credentials via Digital Badges. *New Directions for Community Colleges*, *2020*(189), 39–51.

Summary
"A number of IT firms are partnering with colleges to develop and deliver curriculum and professional credentials that are more continuously tuned to the rapidly changing job market, including embedding various technology certifications and digital badges into degree programs. To illustrate the opportunity for collaboration and innovation between traditional degrees and non-traditional credentials, this chapter provides a case study of IBM, a technology firm with global reach, and a pioneer in this areag iven its significant investments, scale, and success. After developing and scaling its consumer-facing badge offerings, IBM partnered in 2017 with an accredited academic institution, Northeastern University, to articulate these digital badges into degree program credit (Northeastern University,2017)" (p. 39-40).

Method and limitations
"In late 2017, Northeastern University and IBM announced a partner-ship agreement that would allow individuals with an IBM-issued badge to receive graduate credit upon enrolling in select Northeastern professional master degree programs. In doing so, Northeastern University became the first university to recognize IBM digital badge credentials toward graduate degree programs and certificates (Northeastern University, 2017). This partnership emerged from IBM and Northeastern's existing relationship and mutual interest in eliminating the gap between learning and work: IBM is a recognized innovator in new forms of credentialing, and Northeastern in work-integrated learning. While the partnership focused initially on map-ping IBM badges to advanced standing in professional graduate degrees, the same basic strategy could apply to undergraduate degrees and certificates at both 2- and 4-year institutions. Indeed, following a similar approach, Northeastern agreed to recognize Google's IT Support Professional Certificate for academic credit in its IT bachelor degree completion program in2018 (Fain, 2018)" (p. 44).

How this relates to my topic and other things I've read? or What does this mean for my research?
Work integrated learning is an important aspect to maintaining institutional relevance, especially considering the microcredentials and self-guided learning paths available The partnership between Northeastern University and IBM, followed by Google and others demonstrates a successful attempt at creating digital badges which convey the credibility and validity of traditional learning models and allows for networking at several recognized technology vendors.

Other articles I need to locate and review:
VanKleef Conley, N., & Gallagher, S. (2018, June 25).
Gallagher, S. R. (2018).

Topic(s):
- Work-integrated learning

Keyword(s):
- Internship
- Learning quality

Database/ index/ source(s):
- Education Resources Information Center (ERIC)

APA Reference
Philip, J., & Gavrilova Aguilar, M. (2021). Student perceptions of leadership skills necessary for digital transformation. *Journal of Education for Business*, 97 (2), 86-98.

Summary
"Digital transformations help organizations revitalize business processes and customer relationships. Industry professionals and academicians recognize that leadership skills and technology skills are complementary when managing a digital workplace. Using a classroom exercise administered in multiple semesters, we attempted to capture student perceptions of leadership skills and competencies they deem important in leading a company through a digital transformation. Findings revealed that younger generations recognize digital literacy for corporate leaders as a key skill alongside conventional leadership skills" (p. 86).

Method and limitations
"This exercise was conducted in three semesters in an attempt to gather student perceptions of digital trans- formation at various stages of their business education, such as at the undergraduate and graduate levels. The exercise was also conducted prior to a discussion of this topic in class and then after the subject matter was taught. Consistent with SoTL's method of connecting the inquiry to student learning (Felten, 2013), our goal was to identify students' viewpoints on this topic when they were initially unaware (either inside or outside the classroom) of this topic's importance in leadership and after they were taught this topic in class. We further discuss the class exercise conducted in various semesters as Study 1 and Study 2 to demonstrate perception changes. Conducting the exercise in multiple semesters also allowed us to analyze results for different student sample sizes. Study 1 was conducted in an undergraduate leader- ship class at a business school in the U.S. The class consisted of 17 students, who were mostly seniors. Study 2 was conducted in two semesters, where students from MBA leadership courses at the same business school were asked to read the article on digital leadership entitled, How Digital Leadership Is(n't) Different (Kane et al., 2019). Then, they were asked to answer the same survey question as Study 1" (p. 90).

How this relates to my topic and other things I've read? or What does this mean for my research?
Learners perceived value of the skills and content of a course affect satisfaction and effectiveness. This study shows that workforce readiness, specifically in the area of digital transformation, recognize relevant activities and readings and as such are motivated and encouraged to perform.

Other articles I need to locate and review:
Tanniru, M. R. (2018).

Topic(s):
- Learning effectiveness

Keyword(s):
- Digital transformation
- Learning quality

Database/ index/ source(s):
- Education Resources Information Center (ERIC)

APA Reference

Chen, Y., Albert, L. J., & Jensen, S. (2021). Innovation farm: Teaching Artificial Intelligence through gamified social entrepreneurship in an introductory MIS course*. *Decision Sciences Journal of Innovative Education, 20, 43-56.*

Summary

"…we designed and implemented the "Innovation Farm," a gamified social entrepreneurship startup project, to help students learn the concepts and applications of AI in an introductory management information systems (MIS) course. The students work in teams to propose AI-powered products and services targeting real-world social issues. The project cultivates social awareness and empathy in students, fosters innovation, and introduces them to current AI technologies while also reinforcing their understanding of MIS course content. Students found the gamified approach engaging, as they are tackling real-world social issues while also learning about innovation, collaboration, branding, and fundraising for a social startup. Students reported positive outcomes related to innovating using technology, making an impact through real-world projects, and practicing their teamwork and leadership skills" (p. 86). "In general, students responded positively to the learning experience about real-world projects, social impact, and teamwork. They also demonstrated an enhanced level of understanding of AI, social responsibility, and the potential for using technology to make a difference" (p. 54).

Topic(s):

- Learning effectiveness

Keyword(s):

- Artificial intelligence
- Gamification

Database/ index/ source(s):

- Education Resources Information Center (ERIC)

Method and limitations

"The pilot implementation of the Innovation Farm included 96 students enrolled in two sections of the course—48 in each section, respectively. The students formed 17 teams, each consisting of 4 to 6 teammates based on students' topic preferences" (p. 48).
"…the length of time from the beginning of the project to the end may lead to recall difficulties (Pratt et al.,2000), although some studies have found that retrospective responses are not influenced by memory issues (Townsend & Wilton,2003). Further, post-then-pretests do not directly measure learning and thus they are subject to the same limitations as other self-report methods (Pratt et al.,2000; Townsend & Wilton,2003)" (p. 50-52).

How this relates to my topic and other things I've read? or What does this mean for my research?

Gamification is another related educational strategy to engage learners and to improve learning satisfaction and effectiveness. Leveraging social entrepreneurship to model gamification appears to promote awareness as well as classroom collaboration.

Other articles I need to locate and review:

Damani, B., Sardeshpande, V. & Gaitonde, U. (2015)

APA Reference
Wheelahan, L., & Moodie, G. (2021). Analysing micro-credentials in higher education: a Bernsteinian analysis. *Journal of Curriculum Studies*, *53*(2), 212–228.

Summary
"This paper argues that while micro-credentials are an extension of the discourse of employability skills, their potential to 'discipline' the higher education curriculum to align it more closely with putative labour market requirements is more far-reaching than hitherto. This is because micro-credentials weaken relations of classification of knowledge and framing (the pacing, sequencing and evaluation) of knowledge more than other previous innovations in higher education curriculum. The paper provides a framework for analysing micro-credentials. It is primarily situated in the social realist school within the sociology of education, but it also draws from the Continental *Didaktik* tradition. Each can contribute to the other, and in the process make an important contribution to theorizing about the nature of educational knowledge in higher education. Such a dialogue can help to overcome problems of disciplinary fragmentation in theorizing the nature of curriculum and educational knowledge in Anglophone traditions (Biesta, 2007; Furlong & Whitty, 2017)" (p. 213).

Topic(s):
- Learning effectiveness

Keyword(s):
- Microcredentials
- Higher education

Database/ index/ source(s):
- Google Scholar

Method and limitations
"The first section of this paper explains what micro-credentials are, where they come from, and how they have moved to the centre of higher education policy in many countries. The second section analyses why there is a gap in theorizing educational knowledge in higher education to help explain the relative lack of opposition to micro-credentials, and it contrasts Anglophone traditions with *Didaktik* traditions. It draws on the latter to consider questions that should be asked in theorizing about curriculum in higher education. This section then explores the state of theorizing about educational knowledge in higher education to show that the distinction between curriculum on the one hand, and pedagogy or teaching and learning on the other, contributes to instrumental discourses about the purposes of, and outcomes from, higher education. The next section draws on social realism and Basil Bernstein to assemble the tools needed to analyse micro-credentials and their precursors, employability skills and 21st century skills" (p. 213).

How this relates to my topic and other things I've read? or What does this mean for my research?
Comparison of traditional theory knowledge compared and establishing a framework to evaluate microcredentials is important when understanding a learners objectives and preferred learning style.

Other articles I need to locate and review:
Ifenthaler, D., Bellin-Mularski, N., & Mah, D.-K. (Eds.). (2016).
Oliver, B. (2019).

APA Reference
Trongtorsak, S., Saraubon, K., & Nilsook, P. (2021). Collaborative Experiential Learning Process for Enhancing Digital Entrepreneurship. Higher Education Studies, 11(1), 137.

Summary
"The objectives of this research were as follows: 1) to develop the collaborative experiential learning process for enhancing digital entrepreneurship, and 2) to carry out a suitability assessment of this process. In this study, the researcher performed analysis and synthesis involving the process of experiential learning, collaborative learning, and digital entrepreneurship learning, and 21 experts with three years' experience in a related field perform a suitability assessment of the collaborative experiential learning process" (p. 137).

Method and limitations
"The research methodology has two phases, as follows
4.1 Phase I: The Development of a Collaborative Experiential Learning Process for Enhancing Digital Entrepreneurship
Studies of the experiential learning process, collaborative learning process, and digital entrepreneur learning process based upon the information gathered from documents, textbooks, academic journals, concepts, and theories of relevant research papers, including analysis and synthesis of the data using a content analysis approach.
Use the data obtained from the synthesis approach to developing the collaborative experiential learning process for enhancing digital entrepreneurship.
4.2 Phase II: A Suitability Assessment of the Collaborative Experiential Learning Process for Enhancing Digital Entrepreneurship
A questionnaire with five rating scales was employed as a data collection tool. Twenty-one experts with three years' experience in a related field performed the assessment (purposive sampling)" (p. 139).

How this relates to my topic and other things I've read? or What does this mean for my research?
Strategies for instructional design include collaborative processes and other techniques which may affect learning performance and effectiveness.

Other articles I need to locate and review:
Laar, E., van Deursen, A. J., van Dijk, J. A., & de Haan, J. (2018).

Topic(s):
- Instructional design

Keyword(s):
- Instructional design
- Collaborative learning

Database/ index/ source(s):
- Education Resources Information Center (ERIC)

APA Reference

Sprenger, D. A., & Schwaninger, A. (2021). Technology acceptance of four digital learning technologies (classroom response system, classroom chat, e-lectures, and mobile virtual reality) after three months' usage. International Journal of Educational Technology in Higher Education, 18(1).

Summary

"We compared four digital learning technologies (e-lectures, classroom response system, classroom chat, and mobile virtual reality) in terms of their technology acceptance. The classroom response system had the highest level of acceptance. It was closely followed by e-lectures, then the classroom chat and then mobile virtual reality. The students evaluated all tools favorably before and after usage, except for mobile virtual reality, which saw a substantial drop in perceived usefulness and behavioral intention after 3 months' usage" (p. 1).

Method and limitations

"The participants in this study consisted of first term students in the course "General Psychology 1" at the University of Applied Sciences and Arts Northwestern Switzerland. Participation in the study was voluntary. A total of 94 students of the 129 registered students participated (72 females and 22 males). The students completed the survey during class to increase completion rates by providing class time and to reduce potentially distracting activities while filling out the questionnaire (Stieger & Reips, 2010)" (p. 5).

How this relates to my topic and other things I've read? or What does this mean for my research?

While the journal observes that the classroom response has the highest acceptance rating, I was curious how the other educational technologies might augment a classroom response environment and the learning experience.

Other articles I need to locate and review:

Rahman, R. A., Ahmad, S., & Hashim, U. R. (2018).

Topic(s):

- Educational Technologies

Keyword(s):

- Instructional design
- Digital learning

Database/ index/ source(s):

- Education Resources Information Center (ERIC)

APA Reference

Cheng, Z., Richardson, J. C., & Newby, T. J. (2019). Using digital badges as goal-setting facilitators: a multiple case study. *Journal of Computing in Higher Education*, *32*(2), 406–428.

Summary

"Students' goal-setting skills are highly related to their academic learning performance and level of motivation. A review of the literature demonstrated limited research on both applicable goal-setting strategies in higher education and the support of technology in facilitating goal-setting processes. Addressing these two gaps, this study explored the use of digital badges as an innovative approach to facilitate student goal-setting. The digital badge is a digital technology that serves as both a micro-credential and a micro-learning platform. A digital badge is a clickable badge image that represents an accomplished skill or knowledge and includes a variety of metadata such as learning requirements, instructional materials, endorsement information, issue data and institution, which allows the badges to be created, acquired and shared in an online space. In higher education, digital badges have the potential for assisting students by promoting strategic management of the learning process, encouraging persistence and devoted behavior to learning tasks, and improving learning performance" (p. 406).

Topic(s):

- Educational Technologies

Keyword(s):

- Instructional design
- Digital badges

Database/ index/ source(s):

- Education Resources Information Center (ERIC)

Method and limitations

"The two groups of students, one group with high and the other group with low self-efficacy for self-regulated learning were selected as cases in this study. A theoretical replication logic was used to select cases across groups aimed at predicting contrasting results for anticipatable reasons derived from theories and literature. And a literal replication logic was used to identify two identical cases within each group (Yin 2018). In this study, Kevin and Anna (pseudonyms) had high SESR scores while Kate and Liza (pseudonyms) had low SESR (self regulated learning) scores" (p. 411). "Although we included multiple cases and diferent perspectives in the study, semi-interview data was the only source of information, which might decrease the credibility of study. Also, the participants in this study represent a small sample of learners in one tertiary-level context, and thus a generalization of the research fndings to a large population is restricted. However, this exploratory case study provided insights for future research to investigate (1) the relationship between the use of DBs and self-regulated learning, possible mediators include but are not restricted to gender, motivation level, and learning styles; and (2) the relationship between the use of DBs and actual learning performance" (p. 423)

How this relates to my topic and other things I've read? or What does this mean for my research?

Learner satisfaction is closely coupled with being able to identify goals and receiving actionable feedback. This study identifies the role of digital badging and indicates the learner's ability to improving goal commitment.

Other articles I need to locate and review:

Ostashewski, N., & Reid, D. (2015).
Grant, S. (2016).

APA Reference
Basilotta-Gómez-Pablos, V., Matarranz, M., Casado-Aranda, L.-A., & Otto, A. (2022). Teachers' digital competencies in higher education: a systematic literature review. *International Journal of Educational Technology in Higher Education*, *19*(1).

Summary
"The aim of this research is to present a systematic review of the literature in the Web of Science and Scopus, to identify, analyze and classify the published articles between 2000 and 2021 on digital competences, and thus find and improve the research being done on digital skills and future avenues of teachers in the university context" (p. 1).

Method and limitations
"The current study carried out consultations on 10 May 2021 on the WoS and Scopus databases about teachers' digital competence in tertiary/higher education. The starting search revealed over 343 articles in English, 152 of which were duplicates and therefore, were eliminated from further analysis. We then obtained 191 papers published between January 2000 and May 2021. The search was then focused and only those articles that specifically aligned with the research objectives on components of digital competence, ICT resources and educational projects and level of teachers' digital competences were selected. After this filtering, 56 articles were the basis for the bibliometric analysis. In the analysis, we used the SciMAT open-source software (Cobo et al., 2011), which constitutes a useful tool for exploring the theoretical background of a given branch. Considering the output derived from the previous queries, the SciMAT software (i) classified the 56 manuscripts by publication date, citations, and journal titles and (ii) implemented a co-word search to clarify the most relevant topics associated with TDC in higher education. Tis co-word analysis constitutes a content analysis tool that uses patterns of co-occurrence of several items (such as words or nouns) within a collection of manuscripts aiming to recognize the links between ideas within the subject topics in the corpora. In this case, the co-word phase used text-mining tools for the titles, abstracts, and keywords, leading to the development of a strategic diagram highlighting the relative relevance of the topics associated with the competences of higher education teachers" (p. 4).

How this relates to my topic and other things I've read? or What does this mean for my research?
This study identifies the importance and expectations of digital competency with respect to higher education. Furthermore, the study reinforces education teachers need continuously move toward competency development in their educational practices,

Other articles I need to locate and review:
Portillo, J., Garay, U., Tejada, E., & Bilbao, N. (2020).
Santos, C., Pedro, N., & Mattar, J. (2021).
Amhag, L., Hellstrom, L., & Stigmar, M. (2019).

Topic(s):
- Learning Effectiveness

Keyword(s):
- Instructional design
- Digital competence

Database/ index/ source(s):
- Education Resources Information Center (ERIC)